Algorithm for Cholesky Decomposition

Input: an $n \times n$ SPD matrix A
Output: the Cholesky factor, a lower triangular matrix L such that $A = LL^T$
Theorem:(proof omitted) For a symmetric matrix A, the Cholesky algorithm will succeed with non-zero diagonal entries in L if and only if A is SPD. If A is not SPD then the algorithm will either have a zero entry in the diagonal of some L_k (making L_k singular) or will require complex numbers in step 0 or step 1.2.
Notation:
L_{k-1}: the $k-1 \times k-1$ upper left corner of L
a_k: the first $k-1$ entries in column k of A
ℓ_k: the first $k-1$ entries in column k of L^T
a_{kk} and ℓ_{kk}: the kk entries of A and L, respectively.

0) Initialize $L_1 = \sqrt{a_{11}}$
1) For $k = 2, \ldots, n$
1.1) Solve $L_{k-1}\ell_k = a_k$ for ℓ_k (L_{k-1} is $k-1 \times k-1$: for $k = 2$ this is a 1×1 or scalar equation)
1.2) $\ell_{kk} = \sqrt{a_{kk} - \ell_k^T \ell_k}$.
1.3) $L_k = \begin{pmatrix} L_{k-1} & 0 \\ \ell_k^T & \ell_{kk} \end{pmatrix}$.

Cholesky Example

For $A = \begin{pmatrix} 16 & 4 & 4 & -4 \\ 4 & 10 & 4 & 2 \\ 4 & 4 & 6 & -2 \\ -4 & 2 & -2 & 4 \end{pmatrix}$ and $b = \begin{pmatrix} 32 \\ 26 \\ 20 \\ -6 \end{pmatrix}$ construct a Cholesky decompostion and solve $Ax = b$.

Solution:

k = 1: $L_1 = \sqrt{16} = 4$

k=2: $L_1 = 4, a_2 = 4, a_{22} = 10$, Solve the 1×1 system $L_1\ell_2 = a_2$ or $4\ell_2 = 4$ so $\ell_2 = 1$. $\ell_{22} = \sqrt{10-1} = 3$. Therefore $L_2 = \begin{pmatrix} 4 & 0 \\ 1 & 3 \end{pmatrix}$.

k=3: $L_2 = \begin{pmatrix} 4 & 0 \\ 1 & 3 \end{pmatrix}, a_3 = \begin{pmatrix} 4 \\ 4 \end{pmatrix}, a_{33} = 6$. Solve the 2×2 system $L_2\ell_3 = a_3$ or $\begin{pmatrix} 4 & 0 \\ 1 & 3 \end{pmatrix} \ell_3 = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$ so $\ell_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. $\ell_{33} = \sqrt{6 - \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}} = 2$. Therefore $L_3 = \begin{pmatrix} 4 & 0 & 0 \\ 1 & 3 & 0 \\ 1 & 1 & 2 \end{pmatrix}$

k=4: $L_3 = \begin{pmatrix} 4 & 0 & 0 \\ 1 & 3 & 0 \\ 1 & 1 & 2 \end{pmatrix}$, $a_4 = \begin{pmatrix} -4 \\ 2 \\ -2 \end{pmatrix}, a_{44} = 4$. Solve the 3×3 system $L_3\ell_4 = a_4$ or $\begin{pmatrix} 4 & 0 & 0 \\ 1 & 3 & 0 \\ 1 & 1 & 2 \end{pmatrix} \ell_4 = \begin{pmatrix} -4 \\ 2 \\ -2 \end{pmatrix}$ so $\ell_4 = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$. $\ell_{44} = \sqrt{4 - \begin{pmatrix} -1 & 1 & -1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}} = 1$.

Therefore $L = L_4 = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 1 & 2 & 0 \\ -1 & 1 & -1 & 1 \end{pmatrix}$

To solve $Ax = b$ note that $LL^Tx = b$ which can be solved by solving $Lc = b$ and then $L^Tx = c$.
$Lc = b \Rightarrow c = \begin{pmatrix} 8 \\ 6 \\ 3 \\ -1 \end{pmatrix}$ and $L^Tx = c \Rightarrow x = \begin{pmatrix} 1 \\ 2 \\ 1 \\ -1 \end{pmatrix}$.